

Unit 7. Eigenvalues and Eigenvectors

1 Complex roots and the eigenvalue problem

1.1 Complex scalars and finite-dimensional algebra

Definition 7.1 (Complex number, conjugate, and modulus).

$$\mathbb{C} = \{a + ib : a, b \in \mathbb{R}\}, \quad i^2 = -1.$$

For $z = a + ib$ and $w = c + id$,

$$z + w = (a + c) + i(b + d), \quad zw = (ac - bd) + i(ad + bc).$$

The **conjugate**, **real part**, **imaginary part**, and **modulus** of z are

$$\bar{z} = a - ib, \quad \Re z = a, \quad \Im z = b, \quad |z| = (a^2 + b^2)^{1/2}.$$

A real number is identified with $a + i0$.

Origin: A&M §1.2 and Exs. 1.17–1.19

Theorem 7.2 (Complex scalar identities). For $z, w \in \mathbb{C}$,

$$\overline{z + w} = \bar{z} + \bar{w}, \quad \overline{zw} = \bar{z} \bar{w}, \quad \overline{\bar{z}} = z,$$

$$z\bar{z} = |z|^2, \quad |zw| = |z||w|, \quad |\bar{z}| = |z|.$$

Moreover,

$$z \neq 0 \implies z^{-1} = \frac{\bar{z}}{|z|^2}.$$

With the preceding addition and multiplication, $\mathbb{C}$ is a field.

Origin: A&M Exs. 1.17–1.19

Proof. Writing $z = a + ib$ and $w = c + id$ gives

$$\overline{zw} = (ac - bd) - i(ad + bc) = (a - ib)(c - id) = \bar{z} \bar{w},$$

$$z\bar{z} = (a + ib)(a - ib) = a^2 + b^2 = |z|^2,$$

and the remaining conjugation identities follow entrywise. Hence

$$|zw|^2 = zw\overline{zw} = z\bar{z}w\bar{w} = |z|^2|w|^2.$$

Both sides are nonnegative, so $|zw| = |z||w|$. If $z \neq 0$, then $|z|^2 = a^2 + b^2 > 0$ and

$$z \frac{\bar{z}}{|z|^2} = 1.$$

Associativity, commutativity, distributivity, and the identities 0, 1 follow by expanding real and imaginary parts and applying the real field laws. $\square$

Definition 7.3 (Complex vectors and matrices). For positive integers m, n ,

$$\mathbb{C}^n = \{(z_1, \dots, z_n)' : z_j \in \mathbb{C}\}, \quad \mathbb{C}^{m \times n} = \{\mathbf{A} = (a_{ij}) : a_{ij} \in \mathbb{C}\}.$$

Addition, scalar multiplication, matrix multiplication, span, linear independence, basis, dimension, kernel, image, rank, trace, transpose, and block partitioning are defined by the same coordinate formulas as over $\mathbb{R}$. Write $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ when a statement holds over either field. Then

$$\det \mathbf{A} = \sum_{\sigma} \operatorname{sgn}(\sigma) \prod_{i=1}^n a_{i,\sigma(i)}, \quad \mathbf{A} \in \mathbb{F}^{n \times n},$$

where the sum is over the permutations of $\{1, \dots, n\}$.

Origin: S §§1.1–1.3 and §3.1; A&M §1.2

Theorem 7.4 (Field transfer for finite-dimensional algebra). Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. The basis-extension theorem, rank–nullity, row reduction, row rank equals column rank, and the square inverse criterion hold in $\mathbb{F}^n$. Moreover,

$$\dim \mathbb{F}^{m \times n} = mn, \quad \text{rank}(\mathbf{A}') = \text{rank}(\mathbf{A}) \quad (\mathbf{A} \in \mathbb{F}^{m \times n}).$$

The determinant over $\mathbb{F}$ is multilinear and alternating and satisfies

$$\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}, \quad \det(\mathbf{A}') = \det \mathbf{A}, \quad \mathbf{A} \text{adj}(\mathbf{A}) = \text{adj}(\mathbf{A})\mathbf{A} = (\det \mathbf{A})\mathbf{I}.$$

Block-triangular determinant identities and Sylvester’s identity also hold:

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{O} & \mathbf{D} \end{pmatrix} = \det \mathbf{A} \det \mathbf{D}, \quad \det(\mathbf{I}_m - \mathbf{XY}) = \det(\mathbf{I}_n - \mathbf{YX})$$

for $\mathbf{X} \in \mathbb{F}^{m \times n}$ and $\mathbf{Y} \in \mathbb{F}^{n \times m}$.

Origin: S §§1.1–1.3 and §§3.1–3.3

Proof. Every nonzero pivot in $\mathbb{F}$ has an inverse. The elimination algorithm therefore produces echelon form over either field. Pivot columns give a basis of the image; nonzero rows give a basis of the row space; free variables give a basis of the kernel; appending vectors and eliminating their coordinates proves basis extension. For $T : \mathbb{F}^d \rightarrow \mathbb{F}^e$, counting pivots gives

$$\dim \ker T + \dim \operatorname{im} T = \dim \operatorname{dom} T.$$

For every matrix $\mathbf{A}$,

$$\dim \operatorname{row}(\mathbf{A}) = \#\{\text{pivots of } \mathbf{A}\} = \dim \operatorname{col}(\mathbf{A}).$$

The mn matrices having one entry equal to 1 and all other entries zero form a basis of $\mathbb{F}^{m \times n}$. Transposition exchanges row rank and column rank. The earlier proofs of the displayed determinant identities expand only finite sums and products, cancel a nonzero pivot, or multiply unit block factors. The complex scalar identities supply every field operation used in those proofs, so each equality is unchanged over $\mathbb{C}$. $\square$

1.2 Polynomials over the two fields

Definition 7.5 (Polynomial, root, factor, and multiplicity). For $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$,

$$\mathbb{F}[t] = \left\{ \sum_{j=0}^d c_j t^j : d \geq 0, c_j \in \mathbb{F} \right\}.$$

For a nonzero polynomial $p(t) = \sum_{j=0}^d c_j t^j$ with $c_d \neq 0$, $\deg p = d$. A polynomial q **divides** p , written $q \mid p$, if $p = qr$ for some $r \in \mathbb{F}[t]$. A scalar $\lambda \in \mathbb{F}$ is a **root** of p if $p(\lambda) = 0$. Its **multiplicity** is

$$m_p(\lambda) = \max\{r \geq 1 : (t - \lambda)^r \mid p\}.$$

A degree- d polynomial **splits over** $\mathbb{F}$ if it has the form

$$c \prod_{j=1}^d (t - \lambda_j), \quad c, \lambda_1, \dots, \lambda_d \in \mathbb{F}.$$

Origin: H Appendix 21.A, p. 485

Theorem 7.6 (Division, factors, and polynomial identity). Let $p, q \in \mathbb{F}[t]$ and $q \neq 0$.

(a) There are unique $b, r \in \mathbb{F}[t]$ such that

$$p = bq + r, \quad r = 0 \text{ or } \deg r < \deg q.$$

(b) $p(\lambda) = 0$ if and only if $(t - \lambda) \mid p$.

(c) A nonzero polynomial of degree d has at most d distinct roots.

(d) If two polynomials agree at infinitely many scalars, they are equal.

Origin: H Thms. 21.A.1–21.A.3 and 21.A.5

Proof. If $p = 0$, take $b = r = 0$. Let $p \neq 0$, $\deg p = m$, and $\deg q = d$, with leading coefficients a_m, b_d . For nonzero $f, g \in \mathbb{F}[t]$, their leading coefficients have nonzero product, so

$$\deg(fg) = \deg f + \deg g.$$

If $m < d$, take $b = 0, r = p$. If $m \geq d$, then

$$p_1(t) = p(t) - a_m b_d^{-1} t^{m-d} q(t)$$

has degree less than m . Induction on m gives existence. If $p = bq + r = \tilde{b}q + \tilde{r}$, then

$$(b - \tilde{b})q = \tilde{r} - r.$$

A nonzero left side has degree at least d , while a nonzero right side has degree less than d ; hence both vanish.

Division by $t - \lambda$ gives $p(t) = (t - \lambda)b(t) + r$ with constant r . Evaluation at λ gives $r = p(\lambda)$, proving (b). If $p = (t - \lambda_1)q_1$ and $\lambda_2 \neq \lambda_1$ is another root, then

$$0 = p(\lambda_2) = (\lambda_2 - \lambda_1)q_1(\lambda_2) \implies q_1(\lambda_2) = 0.$$

Repeating this argument gives

$$\lambda_1, \dots, \lambda_k \text{ distinct roots} \implies \prod_{j=1}^k (t - \lambda_j) \mid p,$$

so $k \leq \deg p$. The difference of two polynomials agreeing at infinitely many scalars has infinitely many roots; (c) forces that difference to be zero. $\square$

Theorem 7.7 (Factorisation over $\mathbb{C}$). If $p \in \mathbb{C}[t]$ has degree $d \geq 1$ and leading coefficient c_d , then there are $\lambda_1, \dots, \lambda_d \in \mathbb{C}$ such that

$$p(t) = c_d \prod_{j=1}^d (t - \lambda_j).$$

Each root occurs in this list according to its multiplicity.

Origin: Adapted from H §21.1, Eqs. (21.1.10)–(21.1.11), and Thm. 21.A.3

Proof. The fundamental theorem of algebra supplies a root λ_1 . The factor theorem gives

$$p(t) = (t - \lambda_1)p_1(t), \quad \deg p_1 = d - 1.$$

If $d - 1 > 0$, apply the fundamental theorem of algebra to p_1 and repeat. After d steps the remaining factor is the leading coefficient c_d . If λ occurs r times, then

$$p(t) = (t - \lambda)^r h(t), \quad h(\lambda) = c_d \prod_{\lambda_j \neq \lambda} (\lambda - \lambda_j) \neq 0.$$

The factor theorem excludes $(t - \lambda) \mid h$, so $m_p(\lambda) = r$. □

1.3 Eigenvalues and the characteristic polynomial

Definition 7.8 (Spectrum and multiplicities over a field). Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and $\mathbf{A} \in \mathbb{F}^{n \times n}$. A scalar $\lambda \in \mathbb{F}$ is an **eigenvalue** of $\mathbf{A}$ if

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$$

for some **eigenvector** $\mathbf{x} \in \mathbb{F}^n \setminus \{\mathbf{0}\}$. The **eigenspace**, **spectrum**, and **characteristic poly-**

nomial are

$$\mathcal{E}_\lambda(\mathbf{A}) = \ker(\mathbf{A} - \lambda \mathbf{I}_n), \quad \sigma_{\mathbb{F}}(\mathbf{A}) = \{\lambda \in \mathbb{F} : \mathcal{E}_\lambda(\mathbf{A}) \neq \{\mathbf{0}\}\},$$

$$\chi_{\mathbf{A}}(t) = \det(t\mathbf{I}_n - \mathbf{A}).$$

For $\lambda \in \sigma_{\mathbb{F}}(\mathbf{A})$, its **algebraic multiplicity** is $a_{\mathbf{A}}(\lambda) = m_{\chi_{\mathbf{A}}}(\lambda)$ and its **geometric multiplicity** is

$$g_{\mathbf{A}}(\lambda) = \dim \mathcal{E}_\lambda(\mathbf{A}) = n - \text{rank}(\mathbf{A} - \lambda \mathbf{I}_n).$$

For a real matrix, $\sigma_{\mathbb{R}}(\mathbf{A})$ and $\sigma_{\mathbb{C}}(\mathbf{A})$ distinguish eigenpairs in $\mathbb{R}^n$ from eigenpairs obtained when its entries are regarded as complex.

Origin: H §21.1, eqs. (21.1.1)–(21.1.11); A&M §7.1

Theorem 7.9 (Characteristic criterion and coefficients). For $\mathbf{A} \in \mathbb{F}^{n \times n}$ and $\lambda \in \mathbb{F}$,

$$\lambda \in \sigma_{\mathbb{F}}(\mathbf{A}) \iff \chi_{\mathbf{A}}(\lambda) = 0.$$

Moreover,

$$\chi_{\mathbf{A}}(t) = t^n - (\operatorname{tr} \mathbf{A})t^{n-1} + \cdots + (-1)^n \det \mathbf{A}.$$

Origin: H §21.1, eqs. (21.1.2)–(21.1.8); A&M §7.1

Proof.

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x} \text{ for some } \mathbf{x} \neq \mathbf{0} \iff \ker(\lambda\mathbf{I} - \mathbf{A}) \neq \{\mathbf{0}\} \iff \det(\lambda\mathbf{I} - \mathbf{A}) = 0.$$

In the Leibniz expansion of $\det(t\mathbf{I} - \mathbf{A})$, the identity permutation gives

$$\prod_{i=1}^n (t - a_{ii}) = t^n - \left(\sum_{i=1}^n a_{ii} \right) t^{n-1} + \cdots .$$

A nonidentity permutation has at most $n - 2$ fixed points, so every other term has degree at most

$n - 2$. Thus the first two coefficients are 1 and $-\operatorname{tr} \mathbf{A}$. At $t = 0$,

$$\chi_{\mathbf{A}}(0) = \det(-\mathbf{A}) = (-1)^n \det \mathbf{A}.$$



Theorem 7.10 (Complex spectrum, trace, and determinant). Every $\mathbf{A} \in \mathbb{C}^{n \times n}$ has complex eigenvalues $\lambda_1, \dots, \lambda_n$, listed with algebraic multiplicity, and

$$\chi_{\mathbf{A}}(t) = \prod_{j=1}^n (t - \lambda_j), \quad \operatorname{tr} \mathbf{A} = \sum_{j=1}^n \lambda_j, \quad \det \mathbf{A} = \prod_{j=1}^n \lambda_j.$$

Consequently,

$$\mathbf{A} \text{ is singular} \iff 0 \in \sigma_{\mathbb{C}}(\mathbf{A}).$$

Origin: H §21.6, Thm. 21.6.1; A&M Exs. 7.26–7.27

Proof. The factorisation theorem and the leading coefficient of $\chi_{\mathbf{A}}$ give

$$\chi_{\mathbf{A}}(t) = \prod_{j=1}^n (t - \lambda_j) = t^n - \left(\sum_{j=1}^n \lambda_j \right) t^{n-1} + \cdots + (-1)^n \prod_{j=1}^n \lambda_j.$$

Coefficient comparison with Theorem 7.9 gives the trace and determinant identities. The last equivalence follows from $\det \mathbf{A} = \prod_j \lambda_j$ and the determinant criterion. $\square$

Theorem 7.11 (Conjugate pairs for real matrices). If $\mathbf{A} \in \mathbb{R}^{n \times n}$ and $\lambda \in \sigma_{\mathbb{C}}(\mathbf{A})$, then $\bar{\lambda} \in \sigma_{\mathbb{C}}(\mathbf{A})$ and

$$a_{\mathbf{A}}(\bar{\lambda}) = a_{\mathbf{A}}(\lambda).$$

If n is odd, $\sigma_{\mathbb{R}}(\mathbf{A}) \neq \emptyset$.

Origin: A&M Ex. 7.17; S §5.2

Proof. The coefficients of $\chi_{\mathbf{A}}$ are real, so

$$\chi_{\mathbf{A}}(\bar{z}) = \overline{\chi_{\mathbf{A}}(z)}.$$

Thus $\chi_{\mathbf{A}}(\lambda) = 0$ implies $\chi_{\mathbf{A}}(\bar{\lambda}) = 0$. If

$$\chi_{\mathbf{A}}(t) = (t - \lambda)^m q(t), \quad q(\lambda) \neq 0,$$

then

$$\chi_{\mathbf{A}}(t) = \overline{\chi_{\mathbf{A}}(\bar{t})} = (t - \bar{\lambda})^m \overline{q(\bar{t})}, \quad \overline{q(\bar{\lambda})} \neq 0.$$

Hence $a_{\mathbf{A}}(\bar{\lambda}) \geq m$; conjugating again gives the reverse inequality. Nonreal roots occur in distinct pairs with equal multiplicity. If no root were real, n would be even. $\square$

Example 7.1 (A real rotation over two fields). For

$$\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \chi_{\mathbf{R}}(t) = t^2 + 1 = (t - i)(t + i),$$

$$\sigma_{\mathbb{R}}(\mathbf{R}) = \emptyset, \quad \sigma_{\mathbb{C}}(\mathbf{R}) = \{-i, i\}.$$

Complex eigenvectors are

$$\mathbf{R} \begin{pmatrix} 1 \\ -i \end{pmatrix} = i \begin{pmatrix} 1 \\ -i \end{pmatrix}, \quad \mathbf{R} \begin{pmatrix} 1 \\ i \end{pmatrix} = -i \begin{pmatrix} 1 \\ i \end{pmatrix}.$$

Origin: H §21.1, pp. 509–510; A&M Ex. 7.22

2 Similarity and diagonalisation

2.1 Similarity and invariant eigenspaces

Definition 7.12 (Similarity and invariant subspace). Matrices $\mathbf{A}, \mathbf{B} \in \mathbb{F}^{n \times n}$ are **similar over** $\mathbb{F}$ if

$$\mathbf{B} = \mathbf{S}^{-1}\mathbf{A}\mathbf{S}$$

for some nonsingular $\mathbf{S} \in \mathbb{F}^{n \times n}$. A subspace $\mathcal{U} \subseteq \mathbb{F}^n$ is **invariant under** $\mathbf{A}$ if

$$\mathbf{A}\mathcal{U} = \{\mathbf{A}\mathbf{u} : \mathbf{u} \in \mathcal{U}\} \subseteq \mathcal{U}.$$

Origin: H §§21.1 and 21.3; A&M Ex. 7.24

Theorem 7.13 (Similarity invariants). If $\mathbf{B} = \mathbf{S}^{-1}\mathbf{A}\mathbf{S}$, then

$$\chi_{\mathbf{B}} = \chi_{\mathbf{A}}, \quad \text{rank } \mathbf{B} = \text{rank } \mathbf{A}, \quad \text{tr } \mathbf{B} = \text{tr } \mathbf{A}, \quad \det \mathbf{B} = \det \mathbf{A},$$

and, for every $\lambda \in \mathbb{F}$,

$$\mathcal{E}_{\lambda}(\mathbf{B}) = \mathbf{S}^{-1}\mathcal{E}_{\lambda}(\mathbf{A}).$$

Thus algebraic and geometric multiplicities are preserved.

Origin: H Thm. 21.3.1; A&M Ex. 7.24

Proof.

$$t\mathbf{I} - \mathbf{B} = \mathbf{S}^{-1}(t\mathbf{I} - \mathbf{A})\mathbf{S} \implies \chi_{\mathbf{B}}(t) = \det(\mathbf{S}^{-1})\chi_{\mathbf{A}}(t)\det \mathbf{S} = \chi_{\mathbf{A}}(t).$$

The characteristic coefficients give equality of trace and determinant. Multiplication by an invertible matrix preserves the dimension of the image, so $\text{rank } \mathbf{B} = \text{rank } \mathbf{A}$. Finally,

$$\mathbf{B}\mathbf{y} = \lambda\mathbf{y} \iff \mathbf{S}^{-1}\mathbf{A}\mathbf{S}\mathbf{y} = \lambda\mathbf{y} \iff \mathbf{A}(\mathbf{S}\mathbf{y}) = \lambda(\mathbf{S}\mathbf{y}) \iff \mathbf{y} \in \mathbf{S}^{-1}\mathcal{E}_{\lambda}(\mathbf{A}).$$

□

Theorem 7.14 (Geometric multiplicity is bounded by algebraic multiplicity). For every eigenvalue λ of $\mathbf{A} \in \mathbb{F}^{n \times n}$,

$$1 \leq g_{\mathbf{A}}(\lambda) \leq a_{\mathbf{A}}(\lambda).$$

Origin: H Thm. 21.3.4

Proof. Set $g = g_{\mathbf{A}}(\lambda)$. If $g = n$, then $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$ for every $\mathbf{x}$, so $\mathbf{A} = \lambda\mathbf{I}$ and $\chi_{\mathbf{A}}(t) = (t - \lambda)^n$. Let $g < n$. Choose a basis $\mathbf{x}_1, \dots, \mathbf{x}_g$ of $\mathcal{E}_{\lambda}(\mathbf{A})$ and extend it to a basis of $\mathbb{F}^n$. With these basis vectors as the columns of $\mathbf{S}$,

$$\mathbf{S}^{-1}\mathbf{A}\mathbf{S} = \begin{pmatrix} \lambda\mathbf{I}_g & \mathbf{C} \\ \mathbf{0} & \mathbf{D} \end{pmatrix}.$$

Similarity and the block-triangular determinant identity give

$$\chi_{\mathbf{A}}(t) = \det \begin{pmatrix} (t - \lambda)\mathbf{I}_g & -\mathbf{C} \\ \mathbf{0} & t\mathbf{I}_{n-g} - \mathbf{D} \end{pmatrix} = (t - \lambda)^g \chi_{\mathbf{D}}(t).$$

Hence $a_{\mathbf{A}}(\lambda) \geq g \geq 1$. □

Theorem 7.15 (Direct sum of distinct eigenspaces). If $\lambda_1, \dots, \lambda_r$ are distinct eigenvalues of $\mathbf{A}$, then

$$\mathcal{E}_{\lambda_1}(\mathbf{A}) + \dots + \mathcal{E}_{\lambda_r}(\mathbf{A}) = \mathcal{E}_{\lambda_1}(\mathbf{A}) \oplus \dots \oplus \mathcal{E}_{\lambda_r}(\mathbf{A}).$$

In particular, eigenvectors belonging to distinct eigenvalues are linearly independent.

Origin: H Thms. 21.4.1–21.4.2; A&M Ex. 7.31

Proof. For $r = 1$, the assertion is the definition of a one-term sum. Suppose it holds for $r - 1$ and

$$\mathbf{x}_1 + \cdots + \mathbf{x}_r = \mathbf{0}, \quad \mathbf{x}_j \in \mathcal{E}_{\lambda_j}(\mathbf{A}).$$

Applying $\mathbf{A} - \lambda_r \mathbf{I}$ gives

$$\sum_{j=1}^{r-1} (\lambda_j - \lambda_r) \mathbf{x}_j = \mathbf{0}.$$

By the induction hypothesis,

$$(\lambda_j - \lambda_r) \mathbf{x}_j = \mathbf{0}, \quad j < r.$$

The eigenvalues are distinct, so $\mathbf{x}_1 = \cdots = \mathbf{x}_{r-1} = \mathbf{0}$, and the original equality gives $\mathbf{x}_r = \mathbf{0}$. $\square$

2.2 Diagonalisation over the chosen field

Definition 7.16 (Diagonalisation over a field). A matrix $\mathbf{A} \in \mathbb{F}^{n \times n}$ is **diagonalisable over $\mathbb{F}$** if

$$\mathbf{S}^{-1} \mathbf{A} \mathbf{S} = \text{diag}(d_1, \dots, d_n)$$

for some nonsingular $\mathbf{S} \in \mathbb{F}^{n \times n}$. A basis consisting of eigenvectors of one matrix is an **eigenbasis** of that matrix.

Origin: H §21.5a, eq. (21.5.1); A&M Ex. 7.61

Theorem 7.17 (Equivalent diagonalisation criteria). Let $\lambda_1, \dots, \lambda_r$ be the distinct eigenvalues of $\mathbf{A} \in \mathbb{F}^{n \times n}$. Conditions (a)–(d) are equivalent.

- (a) $\mathbf{A}$ is diagonalisable over $\mathbb{F}$.
- (b) $\mathbb{F}^n$ has a basis of eigenvectors of $\mathbf{A}$.
- (c)

$$\mathbb{F}^n = \mathcal{E}_{\lambda_1}(\mathbf{A}) \oplus \cdots \oplus \mathcal{E}_{\lambda_r}(\mathbf{A}).$$

- (d) $\chi_{\mathbf{A}}$ splits into linear factors over $\mathbb{F}$ and

$$g_{\mathbf{A}}(\lambda_j) = a_{\mathbf{A}}(\lambda_j), \quad j = 1, \dots, r.$$

Origin: H Thms. 21.4.3, 21.5.1–21.5.2 and Cor. 21.5.4

Proof. If $\mathbf{S} = (\mathbf{x}_1, \dots, \mathbf{x}_n)$ and $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$, then

$$\mathbf{S}^{-1}\mathbf{A}\mathbf{S} = \mathbf{D} \iff \mathbf{A}\mathbf{S} = \mathbf{S}\mathbf{D} \iff \mathbf{A}\mathbf{x}_j = d_j\mathbf{x}_j, \quad j = 1, \dots, n.$$

Thus (a) and (b) are equivalent. Grouping an eigenbasis by eigenvalue and using the direct-sum theorem proves the equivalence of (b) and (c).

If (c) holds, then

$$n = \sum_{j=1}^r g_{\mathbf{A}}(\lambda_j).$$

The union of bases of the eigenspaces is an eigenbasis, so $\mathbf{A}$ is similar to a diagonal matrix and $\chi_{\mathbf{A}}$ splits. Since

$$g_{\mathbf{A}}(\lambda_j) \leq a_{\mathbf{A}}(\lambda_j), \quad \sum_{j=1}^r a_{\mathbf{A}}(\lambda_j) = n,$$

all inequalities are equalities. Conversely, (d) and the direct-sum theorem give a direct sum of

dimension

$$\sum_{j=1}^r g_{\mathbf{A}}(\lambda_j) = \sum_{j=1}^r a_{\mathbf{A}}(\lambda_j) = n,$$

which is $\mathbb{F}^n$. □

Corollary 7.18 (Distinct eigenvalues suffice). If $\mathbf{A} \in \mathbb{F}^{n \times n}$ has n distinct eigenvalues in $\mathbb{F}$, then it is diagonalisable over $\mathbb{F}$.

Origin: H Cor. 21.5.3; A&M Ex. 7.32

Proof. Choose one eigenvector for each eigenvalue. The distinct-eigenspaces theorem makes the resulting n vectors linearly independent, hence a basis. □

Example 7.2 (A repeated root with one eigenvector).

$$\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \chi_{\mathbf{J}}(t) = (t - 1)^2, \quad \mathcal{E}_1(\mathbf{J}) = \ker \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \operatorname{span} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}.$$

Thus $a_{\mathbf{J}}(1) = 2$, $g_{\mathbf{J}}(1) = 1$, and $\mathbf{J}$ is not diagonalisable over either $\mathbb{R}$ or $\mathbb{C}$.

Origin: H §21.1, pp. 508–509; A&M Ex. 7.33

2.3 Matrix polynomials and related products

Definition 7.19 (Matrix polynomial). For $p(t) = \sum_{j=0}^d c_j t^j \in \mathbb{F}[t]$ and $\mathbf{A} \in \mathbb{F}^{n \times n}$,

$$p(\mathbf{A}) = \sum_{j=0}^d c_j \mathbf{A}^j, \quad \mathbf{A}^0 = \mathbf{I}_n.$$

Origin: H Ex. 21.3; S §3.5

Theorem 7.20 (Polynomial action on eigenvectors). For $p, q \in \mathbb{F}[t]$,

$$(p + q)(\mathbf{A}) = p(\mathbf{A}) + q(\mathbf{A}), \quad (pq)(\mathbf{A}) = p(\mathbf{A})q(\mathbf{A}).$$

If $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$, then

$$p(\mathbf{A})\mathbf{x} = p(\lambda)\mathbf{x}.$$

If $\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$ with $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$, then

$$p(\mathbf{A}) = \mathbf{S} \text{diag}(p(\lambda_1), \dots, p(\lambda_n)) \mathbf{S}^{-1}.$$

If $\mathbf{A}$ is nonsingular, every eigenvalue is nonzero and

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x} \implies \mathbf{A}^{-1}\mathbf{x} = \lambda^{-1}\mathbf{x}.$$

Origin: H Lemma 21.1.3 and Thm. 21.5.1

Proof. If $p(t) = \sum_j c_j t^j$ and $q(t) = \sum_k d_k t^k$, then

$$p(\mathbf{A})q(\mathbf{A}) = \sum_{j,k} c_j d_k \mathbf{A}^{j+k} = (pq)(\mathbf{A}).$$

Also $\mathbf{A}^j \mathbf{x} = \lambda^j \mathbf{x}$ by induction, so

$$p(\mathbf{A})\mathbf{x} = \sum_j c_j \mathbf{A}^j \mathbf{x} = \sum_j c_j \lambda^j \mathbf{x} = p(\lambda)\mathbf{x}.$$

From $\mathbf{A}^j = \mathbf{S}\mathbf{D}^j\mathbf{S}^{-1}$,

$$p(\mathbf{A}) = \mathbf{S}p(\mathbf{D})\mathbf{S}^{-1}.$$

If $\mathbf{A}$ is nonsingular, $\lambda = 0$ is impossible; multiplying $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$ by $\mathbf{A}^{-1}$ and λ^{-1} gives the last identity. $\square$

Theorem 7.21 (Rectangular products). Let $\mathbf{A} \in \mathbb{F}^{m \times n}$ and $\mathbf{B} \in \mathbb{F}^{n \times m}$. Then

$$t^n \chi_{\mathbf{AB}}(t) = t^m \chi_{\mathbf{BA}}(t).$$

Consequently, $\mathbf{AB}$ and $\mathbf{BA}$ have the same nonzero eigenvalues with the same algebraic and geometric multiplicities. For $\lambda \neq 0$, the maps

$$\mathbf{B} : \mathcal{E}_\lambda(\mathbf{AB}) \longrightarrow \mathcal{E}_\lambda(\mathbf{BA}), \quad \lambda^{-1} \mathbf{A} : \mathcal{E}_\lambda(\mathbf{BA}) \longrightarrow \mathcal{E}_\lambda(\mathbf{AB})$$

are inverse linear maps.

Origin: H Thm. 21.10.1; A&M Ex. 7.25; S Ex. 3.14

Proof. For $t \neq 0$, Sylvester's determinant identity gives

$$\begin{aligned} t^n \chi_{\mathbf{AB}}(t) &= t^{m+n} \det(\mathbf{I}_m - t^{-1} \mathbf{AB}) \\ &= t^{m+n} \det(\mathbf{I}_n - t^{-1} \mathbf{BA}) = t^m \chi_{\mathbf{BA}}(t). \end{aligned}$$

The two polynomials agree at infinitely many t , hence agree identically. Their nonzero roots have the same multiplicities.

If $\mathbf{A}\mathbf{B}\mathbf{x} = \lambda\mathbf{x}$, then

$$\mathbf{B}\mathbf{A}(\mathbf{B}\mathbf{x}) = \mathbf{B}(\mathbf{A}\mathbf{B}\mathbf{x}) = \lambda\mathbf{B}\mathbf{x}, \quad \lambda^{-1}\mathbf{A}(\mathbf{B}\mathbf{x}) = \mathbf{x}.$$

Thus $\mathbf{B}\mathbf{x} \neq \mathbf{0}$. The analogous calculation begins with $\mathbf{B}\mathbf{A}\mathbf{y} = \lambda\mathbf{y}$ and gives

$$\mathbf{B}(\lambda^{-1}\mathbf{A}\mathbf{y}) = \mathbf{y}.$$

The two displayed maps are inverse, so their eigenspaces have equal dimension. □

3 Unitary triangularisation and Cayley–Hamilton

3.1 Complex inner products and unitary matrices

Definition 7.22 (Adjoint, complex inner product, and unitary matrix). For $\mathbf{A} = (a_{ij}) \in \mathbb{C}^{m \times n}$, its **conjugate transpose** or **adjoint** is

$$\mathbf{A}^* = \overline{\mathbf{A}}' = (\bar{a}_{ji}) \in \mathbb{C}^{n \times m}.$$

For $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$,

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^* \mathbf{y} = \sum_{j=1}^n \bar{x}_j y_j, \quad \|\mathbf{x}\| = (\mathbf{x}^* \mathbf{x})^{1/2}.$$

A list $\mathbf{q}_1, \dots, \mathbf{q}_k$ is **orthonormal** if $\mathbf{q}_i^* \mathbf{q}_j$ equals 1 for $i = j$ and 0 otherwise. A matrix $\mathbf{Q} \in \mathbb{C}^{n \times n}$ is **unitary** if

$$\mathbf{Q}^* \mathbf{Q} = \mathbf{I}_n.$$

Origin: A&M §1.2, Exs. 1.18–1.19 and 7.34

Theorem 7.23 (Adjoint and unitary identities). For conformable complex matrices and complex vectors,

$$\begin{aligned}
 (\mathbf{A} + \mathbf{B})^* &= \mathbf{A}^* + \mathbf{B}^*, & (\mathbf{AB})^* &= \mathbf{B}^* \mathbf{A}^*, & (\mathbf{A}^*)^* &= \mathbf{A}, \\
 \langle a\mathbf{x} + b\mathbf{y}, \mathbf{z} \rangle &= \bar{a}\langle \mathbf{x}, \mathbf{z} \rangle + \bar{b}\langle \mathbf{y}, \mathbf{z} \rangle, & \langle \mathbf{x}, a\mathbf{y} + b\mathbf{z} \rangle &= a\langle \mathbf{x}, \mathbf{y} \rangle + b\langle \mathbf{x}, \mathbf{z} \rangle, \\
 \langle \mathbf{x}, \mathbf{y} \rangle &= \overline{\langle \mathbf{y}, \mathbf{x} \rangle}, & \mathbf{x}^* \mathbf{x} &= \sum_{j=1}^n |x_j|^2 \geq 0, & \mathbf{x}^* \mathbf{x} = 0 &\iff \mathbf{x} = \mathbf{0}.
 \end{aligned}$$

For square $\mathbf{Q}$, the three conditions below are equivalent:

$$\mathbf{Q}^* \mathbf{Q} = \mathbf{I}, \quad \mathbf{Q}^{-1} = \mathbf{Q}^*, \quad \mathbf{Q} \mathbf{Q}^* = \mathbf{I}.$$

They imply

$$\|\mathbf{Q}\mathbf{x}\| = \|\mathbf{x}\|.$$

Origin: A&M Exs. 1.18–1.19 and 7.34; S §5.1.1

Proof. For the product,

$$[(\mathbf{AB})^*]_{ij} = \overline{\sum_k a_{jk} b_{ki}} = \sum_k \bar{b}_{ki} \bar{a}_{jk} = (\mathbf{B}^* \mathbf{A}^*)_{ij}.$$

The other adjoint and inner-product identities follow from the scalar conjugation identities. Positivity is the displayed sum of nonnegative real numbers.

If $\mathbf{Q}^* \mathbf{Q} = \mathbf{I}$, the square inverse criterion gives $\mathbf{Q}^{-1} = \mathbf{Q}^*$ and hence $\mathbf{Q} \mathbf{Q}^* = \mathbf{I}$; the reverse implications are identical. Finally,

$$\|\mathbf{Q}\mathbf{x}\|^2 = (\mathbf{Q}\mathbf{x})^* (\mathbf{Q}\mathbf{x}) = \mathbf{x}^* \mathbf{Q}^* \mathbf{Q} \mathbf{x} = \mathbf{x}^* \mathbf{x}.$$

□

Theorem 7.24 (Complex Gram–Schmidt and completion). Let $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{C}^n$ be linearly independent. Define

$$\mathbf{u}_1 = \mathbf{v}_1, \quad \mathbf{q}_1 = \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|},$$

$$\mathbf{u}_j = \mathbf{v}_j - \sum_{i=1}^{j-1} (\mathbf{q}_i^* \mathbf{v}_j) \mathbf{q}_i, \quad \mathbf{q}_j = \frac{\mathbf{u}_j}{\|\mathbf{u}_j\|}, \quad j = 2, \dots, k.$$

Then every $\mathbf{u}_j$ is nonzero, $\mathbf{q}_1, \dots, \mathbf{q}_k$ are orthonormal, and

$$\text{span}\{\mathbf{q}_1, \dots, \mathbf{q}_j\} = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_j\}.$$

Every complex orthonormal list extends to an orthonormal basis of $\mathbb{C}^n$.

Origin: Adapted from A&M Ex. 7.34; H Thms. 6.4.1 and 6.4.5

Proof. For $j = 1$, $\mathbf{u}_1 = \mathbf{v}_1 \neq \mathbf{0}$, $\mathbf{q}_1^* \mathbf{q}_1 = 1$, and the two one-vector spans agree. Suppose the claims hold through $j - 1$. If $\mathbf{u}_j = \mathbf{0}$, then

$$\mathbf{v}_j = \sum_{i=1}^{j-1} (\mathbf{q}_i^* \mathbf{v}_j) \mathbf{q}_i \in \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_{j-1}\},$$

contrary to independence. For $\ell < j$,

$$\mathbf{q}_\ell^* \mathbf{u}_j = \mathbf{q}_\ell^* \mathbf{v}_j - \sum_{i=1}^{j-1} (\mathbf{q}_i^* \mathbf{v}_j) \mathbf{q}_\ell^* \mathbf{q}_i = 0.$$

Also $\mathbf{q}_j^* \mathbf{q}_j = 1$, and

$$\mathbf{u}_j \in \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_j\}, \quad \mathbf{v}_j = \mathbf{u}_j + \sum_{i=1}^{j-1} (\mathbf{q}_i^* \mathbf{v}_j) \mathbf{q}_i,$$

which proves both span inclusions.

Extend an orthonormal list to a basis by the field-transfer theorem and apply the displayed recursion. For an original orthonormal vector $\mathbf{q}_j$,

$$\mathbf{q}_j - \sum_{i < j} (\mathbf{q}_i^* \mathbf{q}_j) \mathbf{q}_i = \mathbf{q}_j, \quad \|\mathbf{q}_j\| = 1,$$

so the original list is unchanged. □

3.2 Triangular matrices and Schur's theorem

Theorem 7.25 (Upper-triangular calculus). Let $\mathbf{T} \in \mathbb{F}^{n \times n}$ be upper triangular with diagonal $\tau_1, \dots, \tau_n$.

(a)

$$\chi_{\mathbf{T}}(t) = \prod_{j=1}^n (t - \tau_j).$$

(b) If $\mathbf{U}$ is upper triangular, then $\mathbf{TU}$ is upper triangular and

$$(\mathbf{TU})_{ii} = \tau_i u_{ii}.$$

(c) For $p \in \mathbb{F}[t]$, $p(\mathbf{T})$ is upper triangular with diagonal $p(\tau_1), \dots, p(\tau_n)$.

(d) If every $\tau_i \neq 0$, then $\mathbf{T}^{-1}$ is upper triangular with diagonal $\tau_1^{-1}, \dots, \tau_n^{-1}$.

(e) If $\mathbf{N}$ is strictly upper triangular, then $\mathbf{N}^n = \mathbf{O}$.

Origin: H Cor. 21.2.2; A&M Exs. 7.55–7.56

Proof. Part (a) is the triangular determinant identity. For $i > j$,

$$(\mathbf{T}\mathbf{U})_{ij} = \sum_{k=1}^n t_{ik} u_{kj} = 0,$$

because a nonzero summand would require $i \leq k \leq j$. For $i = j$, only $k = i$ remains. Induction gives the diagonal of every power, hence (c).

If all $\tau_i \neq 0$, triangular substitution solves $\mathbf{T}\mathbf{x} = \mathbf{e}_j$ uniquely. Its first $j - 1$ entries vanish, so column j of $\mathbf{T}^{-1}$ has zeros below its diagonal; the j th pivot gives diagonal entry τ_j^{-1} .

For a strictly upper-triangular $\mathbf{N}$, induction on r gives

$$(\mathbf{N}^r)_{ij} = 0 \quad \text{whenever } j - i < r.$$

For $r = n$, every index pair satisfies $j - i < n$, so $\mathbf{N}^n = \mathbf{O}$. □

Theorem 7.26 (Schur triangularisation). For every $\mathbf{A} \in \mathbb{C}^{n \times n}$ and every ordering $\lambda_1, \dots, \lambda_n$ of

its eigenvalues with algebraic multiplicity, there is a unitary $\mathbf{Q}$ such that

$$\mathbf{Q}^* \mathbf{A} \mathbf{Q} = \mathbf{T},$$

where $\mathbf{T}$ is upper triangular with diagonal $\lambda_1, \dots, \lambda_n$. If $\mathbf{A}$ is real and $\chi_{\mathbf{A}}$ splits over $\mathbb{R}$, then $\mathbf{Q}$ and $\mathbf{T}$ may be chosen real, with $\mathbf{Q}'\mathbf{Q} = \mathbf{I}$.

Origin: S Thm. 5.1; H Thm. 21.5.11; A&M Ex. 7.62

Proof. Induct on n . For $n = 1$, take $\mathbf{Q} = (1)$ and $\mathbf{T} = \mathbf{A}$. Choose $\mathbf{x} \in \mathcal{E}_{\lambda_1}(\mathbf{A}) \setminus \{\mathbf{0}\}$, set $\mathbf{q}_1 = \mathbf{x}/\|\mathbf{x}\|$, and complete $\mathbf{q}_1$ to an orthonormal basis:

$$\mathbf{Q}_1 = (\mathbf{q}_1, \mathbf{V}), \quad \mathbf{Q}_1^* \mathbf{Q}_1 = \mathbf{I}.$$

Set $\mathbf{B} = \mathbf{V}^* \mathbf{A} \mathbf{V}$. Since $\mathbf{A} \mathbf{q}_1 = \lambda_1 \mathbf{q}_1$ and $\mathbf{V}^* \mathbf{q}_1 = \mathbf{0}$,

$$\mathbf{Q}_1^* \mathbf{A} \mathbf{Q}_1 = \begin{pmatrix} \lambda_1 & \mathbf{q}_1^* \mathbf{A} \mathbf{V} \\ \mathbf{0} & \mathbf{B} \end{pmatrix}.$$

Similarity and the block-triangular determinant identity give

$$\chi_{\mathbf{A}}(t) = (t - \lambda_1) \chi_{\mathbf{B}}(t).$$

Thus $\mathbf{B}$ has the remaining ordered roots $\lambda_2, \dots, \lambda_n$. By induction, a unitary $\mathbf{W}$ satisfies $\mathbf{W}^* \mathbf{B} \mathbf{W} = \mathbf{T}_2$, upper triangular in that order. Set

$$\mathbf{Q} = \mathbf{Q}_1 \operatorname{diag}(1, \mathbf{W}).$$

Then $\mathbf{Q}$ is unitary and

$$\mathbf{Q}^* \mathbf{A} \mathbf{Q} = \begin{pmatrix} \lambda_1 & \mathbf{q}_1^* \mathbf{A} \mathbf{V} \mathbf{W} \\ \mathbf{0} & \mathbf{T}_2 \end{pmatrix}.$$

For real $\mathbf{A}$ with real roots, choose a real eigenvector and use real orthonormal completion at every induction step. $\square$

Theorem 7.27 (Polynomial spectral mapping). Let $\lambda_1, \dots, \lambda_n$ be the complex eigenvalues of $\mathbf{A} \in \mathbb{C}^{n \times n}$, counted with algebraic multiplicity. For every $p \in \mathbb{C}[t]$,

$$\chi_{p(\mathbf{A})}(t) = \prod_{j=1}^n (t - p(\lambda_j)).$$

Thus

$$\sigma_{\mathbb{C}}(p(\mathbf{A})) = p(\sigma_{\mathbb{C}}(\mathbf{A})),$$

with algebraic multiplicities combined when distinct λ_j have the same image under p .

Origin: H Lemma 21.1.3 and Thm. 21.5.11; S Prop. 5.7; A&M Ex. 7.65

Proof. Choose a Schur factorisation $\mathbf{A} = \mathbf{Q}\mathbf{T}\mathbf{Q}^*$. Matrix-polynomial algebra gives

$$p(\mathbf{A}) = \mathbf{Q}p(\mathbf{T})\mathbf{Q}^*.$$

The triangular-calculus theorem makes $p(\mathbf{T})$ upper triangular with diagonal $p(\lambda_1), \dots, p(\lambda_n)$. Similarity and the triangular characteristic polynomial give the stated factorisation. $\square$

3.3 Cayley–Hamilton and power reduction

Theorem 7.28 (Cayley–Hamilton). For $\mathbf{A} \in \mathbb{F}^{n \times n}$, if

$$\chi_{\mathbf{A}}(t) = t^n + c_{n-1}t^{n-1} + \dots + c_1t + c_0,$$

then

$$\chi_{\mathbf{A}}(\mathbf{A}) = \mathbf{A}^n + c_{n-1}\mathbf{A}^{n-1} + \cdots + c_1\mathbf{A} + c_0\mathbf{I} = \mathbf{O}.$$

Origin: H Ex. 21.3; S Thm. 3.3

Proof. Every entry of $\text{adj}(t\mathbf{I} - \mathbf{A})$ is a polynomial of degree at most $n - 1$. Write

$$\text{adj}(t\mathbf{I} - \mathbf{A}) = \mathbf{B}_0 t^{n-1} + \mathbf{B}_1 t^{n-2} + \cdots + \mathbf{B}_{n-1}.$$

The adjugate identity gives the polynomial identity

$$(t\mathbf{I} - \mathbf{A})(\mathbf{B}_0 t^{n-1} + \cdots + \mathbf{B}_{n-1}) = \chi_{\mathbf{A}}(t)\mathbf{I}.$$

Coefficient comparison gives

$$\begin{aligned} \mathbf{B}_0 &= \mathbf{I}, \\ \mathbf{B}_j - \mathbf{A}\mathbf{B}_{j-1} &= c_{n-j}\mathbf{I}, \quad j = 1, \dots, n-1, \\ -\mathbf{A}\mathbf{B}_{n-1} &= c_0\mathbf{I}. \end{aligned}$$

The recursion yields

$$\mathbf{B}_j = \mathbf{A}^j + c_{n-1}\mathbf{A}^{j-1} + \cdots + c_{n-j}\mathbf{I}.$$

Substitution of $j = n - 1$ into the constant-coefficient equality gives

$$\mathbf{A}^n + c_{n-1}\mathbf{A}^{n-1} + \cdots + c_1\mathbf{A} + c_0\mathbf{I} = \mathbf{O}.$$

□

Corollary 7.29 (Reduction of powers and inverses). For every $p \in \mathbb{F}[t]$ there is a unique remainder r with

$$r = 0 \text{ or } \deg r < n, \quad p(\mathbf{A}) = r(\mathbf{A}),$$

where r is the remainder on division of p by $\chi_{\mathbf{A}}$. If $\mathbf{A}$ is nonsingular, then $c_0 = (-1)^n \det \mathbf{A} \neq 0$ and

$$\mathbf{A}^{-1} = -c_0^{-1} (\mathbf{A}^{n-1} + c_{n-1}\mathbf{A}^{n-2} + \cdots + c_2\mathbf{A} + c_1\mathbf{I}).$$

Origin: H Ex. 21.4; S Ex. 3.22

Proof. The division algorithm gives $p = q\chi_{\mathbf{A}} + r$, $\deg r < n$. Matrix-polynomial algebra and Cayley–Hamilton give

$$p(\mathbf{A}) = q(\mathbf{A})\chi_{\mathbf{A}}(\mathbf{A}) + r(\mathbf{A}) = r(\mathbf{A}).$$

For a nonsingular matrix, $c_0 \neq 0$. Regrouping Cayley–Hamilton as

$$\mathbf{A} \left(\mathbf{A}^{n-1} + c_{n-1}\mathbf{A}^{n-2} + \cdots + c_2\mathbf{A} + c_1\mathbf{I} \right) = -c_0\mathbf{I}$$

and multiplying by $-c_0^{-1}$ gives the inverse formula. □

4 Computation: Numerical Checks in R

4.1 A real matrix with a complex spectrum

$$\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} 1 \\ -i \end{pmatrix}, \quad \mathbf{R}\mathbf{x} = i\mathbf{x}.$$

```

1 Rmat <- matrix(c(0, 1, -1, 0), 2, 2)
2 x <- c(1, -1i)
3 lam_R <- eigen(Rmat)$values
4 c(eigenpair_residual = sqrt(sum(Mod(Rmat %*% x - 1i * x)^2)),
5   trace_gap = Mod(sum(lam_R) - sum(diag(Rmat))),
6   determinant_gap = Mod(prod(lam_R) - det(Rmat)),
7   largest_imaginary_part = max(abs(Im(lam_R))))

```

```

##      eigenpair_residual      trace_gap
##              0              0
##      determinant_gap largest_imaginary_part
##              0              1

```

4.2 Rectangular products and their extra zeros

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 1 & -1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}.$$

```
1 A <- matrix(c(1, 0, 1, 2, 1, -1), 3, 2)
2 B <- matrix(c(2, 1, 1, 1, 0, 0), 2, 3)
3 lam_AB <- eigen(A %*% B)$values
4 lam_BA <- eigen(B %*% A)$values
5 nz_AB <- lam_AB[Mod(lam_AB) > 1e-10]
6 nz_BA <- lam_BA[Mod(lam_BA) > 1e-10]
7 nz_AB <- nz_AB[order(Re(nz_AB), Im(nz_AB))]
8 nz_BA <- nz_BA[order(Re(nz_BA), Im(nz_BA))]
9 c(nonzero_spectrum_gap = max(Mod(nz_AB - nz_BA)),
10   smallest_modulus_AB = min(Mod(lam_AB)),
11   determinant_BA = det(B %*% A))
```

```
## nonzero_spectrum_gap  smallest_modulus_AB
##           8.882e-16           0.000e+00
##           determinant_BA
```

1.000e+00

4.3 A constructed unitary triangularisation

$$\mathbf{A} = \mathbf{Q}\mathbf{T}\mathbf{Q}^*, \quad \mathbf{T} = \begin{pmatrix} 1+2i & 2-i & 1 \\ 0 & -2 & 3+i \\ 0 & 0 & 3-i \end{pmatrix}.$$

```

1 Z <- matrix(c(1+1i, 2-1i, 0+2i,
2               2+0i, -1+1i, 1-2i,
3               0-1i, 1+0i, 2+1i), 3, 3)
4 Q <- qr.Q(qr(Z, LAPACK = TRUE))
5 Tmat <- matrix(0+0i, 3, 3)
6 diag(Tmat) <- c(1+2i, -2+0i, 3-1i)
7 Tmat[1, 2] <- 2-1i
8 Tmat[1, 3] <- 1
9 Tmat[2, 3] <- 3+1i
10 Ac <- Q %*% Tmat %*% Conj(t(Q))
11 That <- Conj(t(Q)) %*% Ac %*% Q
12 ordc <- function(z) order(round(Re(z), 10), round(Im(z), 10))
13 lam_Ac <- eigen(Ac)$values

```

```

14 c(unitarity_residual = norm(Conj(t(Q)) %*% Q - diag(3), "F"),
15    reconstruction_residual = norm(Ac - Q %*% That %*% Conj(t(Q)), "F"),
16    lower_triangle_residual = sqrt(sum(Mod(That[lower.tri(That)])^2)),
17    spectrum_gap =
18      max(Mod(lam_Ac[ordc(lam_Ac)] - diag(Tmat)[ordc(diag(Tmat))]))))

```

```

##          unitarity_residual reconstruction_residual
##          9.683e-16          5.318e-15
## lower_triangle_residual          spectrum_gap
##          6.163e-16          1.601e-15

```

4.4 Cayley–Hamilton on a recurrence matrix

$$\mathbf{C} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -1 & 2 \end{pmatrix}, \quad \chi_{\mathbf{C}}(t) = t^3 - 2t^2 + t - 4.$$

```

1 Cmat <- matrix(c(0, 0, 4, 1, 0, -1, 0, 1, 2), 3, 3)
2 I3 <- diag(3)
3 pC <- Cmat %*% Cmat %*% Cmat -
4      2 * Cmat %*% Cmat + Cmat - 4 * I3

```

```

5 c(cayley_hamilton_residual = norm(pC, "F"),
6   inverse_formula_residual =
7   norm(solve(Cmat) - 0.25 * (Cmat %*% Cmat - 2 * Cmat + I3), "F"))

```

```

## cayley_hamilton_residual inverse_formula_residual
##                        0                        0

```

4.5 Algebraic and geometric multiplicity

$$\mathbf{I}_3, \quad \mathbf{J} = \mathbf{I}_3 + \mathbf{e}_1 \mathbf{e}'_2 + \mathbf{e}_2 \mathbf{e}'_3$$

have the same characteristic polynomial $(t - 1)^3$.

```

1 I3 <- diag(3)
2 J3 <- I3
3 J3[1, 2] <- 1
4 J3[2, 3] <- 1
5 rank_tol <- function(M, tol = 1e-10) qr(M, tol = tol)$rank
6 c(tolerance = 1e-10,
7   rank_I3 = rank_tol(I3),
8   rank_J3 = rank_tol(J3),
9   geometric_multiplicity_I3 = 3 - rank_tol(I3 - diag(3)),
10  geometric_multiplicity_J3 = 3 - rank_tol(J3 - diag(3)),

```

```
11 trace_gap = sum(diag(I3)) - sum(diag(J3)),
12 determinant_gap = det(I3) - det(J3))
```

```
##          tolerance          rank_I3
##          1e-10          3e+00
##          rank_J3 geometric_multiplicity_I3
##          3e+00          3e+00
## geometric_multiplicity_J3          trace_gap
##          1e+00          0e+00
##          determinant_gap
##          0e+00
```

5 Pitfalls

5.1 Omitting the scalar field

$$\sigma_{\mathbb{R}} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \emptyset, \quad \sigma_{\mathbb{C}} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \{-i, i\}.$$

A real matrix has a complex spectrum, but need not have a real eigenpair. Diagonalisation must specify its scalar field.

5.2 Replacing similarity by a list of invariants

$$\mathbf{I}_2, \quad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} : \quad \text{rank} = 2, \quad \text{tr} = 2, \quad \det = 1, \quad \chi(t) = (t - 1)^2.$$

Their eigenspace dimensions are 2 and 1, so they are not similar.

5.3 Comparing eigenvectors entry by entry

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}, \quad c \neq 0 \quad \implies \quad \mathbf{A}(c\mathbf{x}) = \lambda(c\mathbf{x}).$$

For complex eigenvectors $\mathbf{x}, \mathbf{y}$ with $\|\mathbf{x}\| = \|\mathbf{y}\| = 1$, the quantity $|\mathbf{x}^* \mathbf{y}|$ is unchanged when either

vector is multiplied by a scalar of modulus 1. Moreover,

$$\begin{aligned}\|\mathbf{y} - (\mathbf{x}^* \mathbf{y}) \mathbf{x}\|^2 &= \mathbf{y}^* \mathbf{y} - (\mathbf{x}^* \mathbf{y})(\mathbf{y}^* \mathbf{x}) - \overline{\mathbf{x}^* \mathbf{y}}(\mathbf{x}^* \mathbf{y}) + |\mathbf{x}^* \mathbf{y}|^2 \mathbf{x}^* \mathbf{x} \\ &= 1 - |\mathbf{x}^* \mathbf{y}|^2,\end{aligned}$$

so $|\mathbf{x}^* \mathbf{y}| = 1$ exactly when they span the same line.

5.4 Confusing eigenvector identities with matrix identities

$$\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad (\mathbf{J} - \mathbf{I}) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \mathbf{0}, \quad \mathbf{J} - \mathbf{I} \neq \mathbf{0}.$$

An identity on one eigenspace is not an identity on the whole space. Cayley–Hamilton is established by coefficient recursion, not by testing only the available eigenvectors.

6 Exercises

Exercise 7.1. Let $\mathbf{A} \in \mathbb{C}^{n \times n}$, $\mathbf{B} \in \mathbb{C}^{k \times k}$, and $\mathbf{X} \in \mathbb{C}^{n \times k}$ satisfy

$$\mathbf{A}\mathbf{X} = \mathbf{X}\mathbf{B}, \quad \text{rank } \mathbf{X} = k.$$

Prove that $\text{col}(\mathbf{X})$ is invariant under $\mathbf{A}$. Construct a unitary $\mathbf{Q} = (\mathbf{Q}_1, \mathbf{Q}_2)$ such that

$$\mathbf{Q}^* \mathbf{A} \mathbf{Q} = \begin{pmatrix} \mathbf{T}_{11} & \mathbf{T}_{12} \\ \mathbf{0} & \mathbf{T}_{22} \end{pmatrix}, \quad \mathbf{T}_{11} \text{ is similar to } \mathbf{B}.$$

If $k = n$, the second block row and column are omitted. Deduce that every eigenvalue of $\mathbf{B}$ is an eigenvalue of $\mathbf{A}$, with at least the same algebraic multiplicity.

Origin: H Exs. 21.2 and 21.9

Solution. For $\mathbf{u} = \mathbf{X}\mathbf{c}$,

$$\mathbf{A}\mathbf{u} = \mathbf{A}\mathbf{X}\mathbf{c} = \mathbf{X}\mathbf{B}\mathbf{c} \in \text{col}(\mathbf{X}).$$

Write $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_k)$. Apply complex Gram–Schmidt to these columns. It gives

$$\mathbf{X} = \mathbf{Q}_1 \mathbf{R}, \quad \mathbf{Q}_1^* \mathbf{Q}_1 = \mathbf{I}_k,$$

where

$$\mathbf{R} = \mathbf{Q}_1^* \mathbf{X}, \quad r_{ij} = \mathbf{q}_i^* \mathbf{x}_j = 0 \quad (i > j), \quad r_{jj} = \|\mathbf{u}_j\| > 0.$$

Thus $\mathbf{R}$ is nonsingular and upper triangular. Extend the columns of $\mathbf{Q}_1$ to a unitary $\mathbf{Q} = (\mathbf{Q}_1, \mathbf{Q}_2)$ when $k < n$; when $k = n$, set $\mathbf{Q} = \mathbf{Q}_1$ and omit the second block. In either case,

$$\mathbf{A} \mathbf{Q}_1 = \mathbf{A} \mathbf{X} \mathbf{R}^{-1} = \mathbf{X} \mathbf{B} \mathbf{R}^{-1} = \mathbf{Q}_1 \mathbf{R} \mathbf{B} \mathbf{R}^{-1}.$$

Consequently,

$$\mathbf{T}_{11} = \mathbf{Q}_1^* \mathbf{A} \mathbf{Q}_1 = \mathbf{R} \mathbf{B} \mathbf{R}^{-1}.$$

For $k < n$, invariance gives $\mathbf{Q}_2^* \mathbf{A} \mathbf{Q}_1 = \mathbf{O}$ and hence the stated block form, with

$$\chi_{\mathbf{A}}(t) = \chi_{\mathbf{T}_{11}}(t) \chi_{\mathbf{T}_{22}}(t) = \chi_{\mathbf{B}}(t) \chi_{\mathbf{T}_{22}}(t),$$

so $\chi_{\mathbf{B}} \mid \chi_{\mathbf{A}}$. For $k = n$,

$$\mathbf{A} = \mathbf{Q}_1 \mathbf{R} \mathbf{B} \mathbf{R}^{-1} \mathbf{Q}_1^*, \quad \chi_{\mathbf{A}} = \chi_{\mathbf{B}}.$$

Both cases give the multiplicity assertion. 

Exercise 7.2. Let $\mathbf{X} \in \mathbb{C}^{m \times n}$.

(a) Prove

$$\ker(\mathbf{X}^* \mathbf{X}) = \ker \mathbf{X}, \quad \ker(\mathbf{X} \mathbf{X}^*) = \ker \mathbf{X}^*.$$

(b) For $\lambda \neq 0$, prove that

$$\mathbf{X} : \mathcal{E}_{\lambda}(\mathbf{X}^* \mathbf{X}) \longrightarrow \mathcal{E}_{\lambda}(\mathbf{X} \mathbf{X}^*)$$

is an isomorphism with inverse $\lambda^{-1} \mathbf{X}^*$.

(c) Deduce

$$\text{rank}(\mathbf{X}^* \mathbf{X}) = \text{rank} \mathbf{X} = \text{rank}(\mathbf{X} \mathbf{X}^*).$$

Origin: A&M Ex. 7.25; S Ex. 3.14

Solution. For $\mathbf{v} \in \mathbb{C}^n$,

$$\mathbf{X}^* \mathbf{X} \mathbf{v} = \mathbf{0} \implies 0 = \mathbf{v}^* \mathbf{X}^* \mathbf{X} \mathbf{v} = \|\mathbf{X} \mathbf{v}\|^2 \implies \mathbf{X} \mathbf{v} = \mathbf{0}.$$

The reverse implication follows by multiplication by $\mathbf{X}^*$. Replacing $\mathbf{X}$ by $\mathbf{X}^*$ gives the second kernel equality.

If $\mathbf{X}^* \mathbf{X} \mathbf{v} = \lambda \mathbf{v}$, then

$$\mathbf{X} \mathbf{X}^* (\mathbf{X} \mathbf{v}) = \mathbf{X} (\mathbf{X}^* \mathbf{X} \mathbf{v}) = \lambda \mathbf{X} \mathbf{v}, \quad \lambda^{-1} \mathbf{X}^* (\mathbf{X} \mathbf{v}) = \mathbf{v}.$$

Thus $\mathbf{X} \mathbf{v} \neq \mathbf{0}$. Starting from $\mathbf{X} \mathbf{X}^* \mathbf{u} = \lambda \mathbf{u}$ gives

$$\mathbf{X} (\lambda^{-1} \mathbf{X}^* \mathbf{u}) = \mathbf{u},$$

so the maps are inverse. Rank–nullity and part (a) give

$$\text{rank}(\mathbf{X}^* \mathbf{X}) = n - \dim \ker(\mathbf{X}^* \mathbf{X}) = n - \dim \ker \mathbf{X} = \text{rank} \mathbf{X}.$$

Apply the same equality to $\mathbf{X}^*$ and use

$$\text{rank} \mathbf{X}^* = \text{rank} \overline{\mathbf{X}}' = \text{rank} \overline{\mathbf{X}} = \text{rank} \mathbf{X}.$$

Transposition preserves rank; conjugating a column relation conjugates its coefficients and preserves its dimension. ■

Exercise 7.3. Let $\mathbf{u}, \mathbf{v} \in \mathbb{C}^n$, $n \geq 1$. Prove

$$\chi_{\mathbf{u}\mathbf{v}^*}(t) = t^{n-1}(t - \mathbf{v}^*\mathbf{u}).$$

Deduce

$$\det(\mathbf{I} + \mathbf{u}\mathbf{v}^*) = 1 + \mathbf{v}^*\mathbf{u}.$$

If $1 + \mathbf{v}^*\mathbf{u} \neq 0$, prove

$$(\mathbf{I} + \mathbf{u}\mathbf{v}^*)^{-1} = \mathbf{I} - \frac{\mathbf{u}\mathbf{v}^*}{1 + \mathbf{v}^*\mathbf{u}}.$$

Origin: A&M Ex. 7.36; S §3.8, Lemma 2 and eq. (3.14)

Solution. Regard $\mathbf{u}$ as an $n \times 1$ matrix and $\mathbf{v}^*$ as a $1 \times n$ matrix. The rectangular-product identity gives

$$t\chi_{\mathbf{u}\mathbf{v}^*}(t) = t^n\chi_{\mathbf{v}^*\mathbf{u}}(t) = t^n(t - \mathbf{v}^*\mathbf{u}).$$

Cancellation of the common polynomial factor t gives the first identity. Polynomial spectral mapping with $p(t) = 1 + t$ gives

$$\det(\mathbf{I} + \mathbf{u}\mathbf{v}^*) = 1^{n-1}(1 + \mathbf{v}^*\mathbf{u}).$$

Set $\alpha = (1 + \mathbf{v}^*\mathbf{u})^{-1}$. Then

$$\begin{aligned}(\mathbf{I} + \mathbf{u}\mathbf{v}^*)(\mathbf{I} - \alpha\mathbf{u}\mathbf{v}^*) &= \mathbf{I} + [1 - \alpha - \alpha(\mathbf{v}^*\mathbf{u})]\mathbf{u}\mathbf{v}^* \\ &= \mathbf{I}.\end{aligned}$$

The square inverse criterion gives the asserted inverse. ■

Exercise 7.4. For $n \geq 2$, set

$$\mathbf{A} = \mathbf{I}_n, \quad \mathbf{B} = \mathbf{I}_n + \mathbf{e}_1\mathbf{e}_2'.$$

Prove that $\mathbf{A}$ and $\mathbf{B}$ have the same rank, determinant, trace, and characteristic polynomial, but are not similar.

Origin: H Exs. 21.7–21.8; A&M Ex. 7.33

Solution. Both matrices are upper triangular with diagonal entries 1, so

$$\chi_{\mathbf{A}}(t) = \chi_{\mathbf{B}}(t) = (t - 1)^n, \quad \det \mathbf{A} = \det \mathbf{B} = 1, \quad \operatorname{tr} \mathbf{A} = \operatorname{tr} \mathbf{B} = n.$$

Their nonzero determinants give rank n . If $\mathbf{B} = \mathbf{S}^{-1}\mathbf{A}\mathbf{S}$, then

$$\mathbf{B} = \mathbf{S}^{-1}\mathbf{I}_n\mathbf{S} = \mathbf{I}_n,$$

contrary to $\mathbf{e}_1\mathbf{e}'_2 \neq \mathbf{O}$. 

Exercise 7.5. For $\lambda, \mu, a \in \mathbb{F}$, classify the diagonalisability over $\mathbb{F}$ of

$$\mathbf{A} = \begin{pmatrix} \lambda & a \\ 0 & \mu \end{pmatrix}.$$

When it is diagonalisable, give an eigenbasis.

Origin: H Ex. 21.16; A&M Exs. 7.32–7.33

Solution. If $\lambda \neq \mu$, then

$$\mathbf{A} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{A} \begin{pmatrix} a \\ \mu - \lambda \end{pmatrix} = \mu \begin{pmatrix} a \\ \mu - \lambda \end{pmatrix}.$$

The determinant of the matrix formed from these vectors is $\mu - \lambda \neq 0$, so they are an eigenbasis.

If $\lambda = \mu$, then

$$\mathcal{E}_\lambda(\mathbf{A}) = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} : ax_2 = 0 \right\}.$$

For $a = 0$, $\mathbf{A} = \lambda \mathbf{I}_2$ and every basis is an eigenbasis. For $a \neq 0$, the eigenspace has dimension $1 < 2$, so $\mathbf{A}$ is not diagonalisable. ■

Exercise 7.6. Suppose $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$ and

$$\mathbf{y}^* \mathbf{A} = \mu \mathbf{y}^*, \quad \mathbf{x}, \mathbf{y} \neq \mathbf{0}.$$

(a) Prove $\mathbf{A}^* \mathbf{y} = \bar{\mu} \mathbf{y}$.

(b) Prove $\lambda \neq \mu \implies \mathbf{y}^* \mathbf{x} = 0$.

(c) If $\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$, let $\mathbf{s}_j$ be column j of $\mathbf{S}$ and $\mathbf{r}_j^*$ row j of $\mathbf{S}^{-1}$. Prove

$$\mathbf{A}\mathbf{s}_j = d_j\mathbf{s}_j, \quad \mathbf{r}_j^*\mathbf{A} = d_j\mathbf{r}_j^*, \quad \mathbf{r}_i^*\mathbf{s}_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Origin: A&M Ex. 7.37; H Ex. 21.14

Solution. Taking adjoints gives

$$\mathbf{A}^*\mathbf{y} = (\mathbf{y}^*\mathbf{A})^* = (\mu\mathbf{y}^*)^* = \bar{\mu}\mathbf{y}.$$

Also

$$\mu\mathbf{y}^*\mathbf{x} = \mathbf{y}^*\mathbf{A}\mathbf{x} = \lambda\mathbf{y}^*\mathbf{x},$$

so $(\lambda - \mu)\mathbf{y}^*\mathbf{x} = 0$.

The identities $\mathbf{A}\mathbf{S} = \mathbf{S}\mathbf{D}$ and $\mathbf{S}^{-1}\mathbf{A} = \mathbf{D}\mathbf{S}^{-1}$ give the first two column and row equations. The (i, j) entry of $\mathbf{S}^{-1}\mathbf{S} = \mathbf{I}$ is 1 for $i = j$ and 0 otherwise, which gives the last identity. ■

Exercise 7.7. Let $\mathbf{A}$ be diagonalisable over $\mathbb{C}$ with distinct eigenvalues $\lambda_1, \dots, \lambda_r$. Define

$$\ell_j(t) = \prod_{\substack{1 \leq k \leq r \\ k \neq j}} \frac{t - \lambda_k}{\lambda_j - \lambda_k}, \quad \mathbf{P}_j = \ell_j(\mathbf{A}),$$

where an empty product equals 1. Prove

$$\mathbf{P}_j^2 = \mathbf{P}_j, \quad \mathbf{P}_j \mathbf{P}_k = \mathbf{O} \ (j \neq k), \quad \sum_{j=1}^r \mathbf{P}_j = \mathbf{I},$$

$$\text{im } \mathbf{P}_j = \mathcal{E}_{\lambda_j}(\mathbf{A}), \quad \mathbf{A} = \sum_{j=1}^r \lambda_j \mathbf{P}_j.$$

Origin: Adapted from S Prop. 3.19

Solution. Let $\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$, with diagonal entries of $\mathbf{D}$ drawn from $\lambda_1, \dots, \lambda_r$. Since

$$\ell_j(\lambda_k) = \delta_{jk},$$

$\ell_j(\mathbf{D})$ is diagonal with entries 0 or 1. Therefore

$$\ell_j(\mathbf{D})^2 = \ell_j(\mathbf{D}), \quad \ell_j(\mathbf{D})\ell_k(\mathbf{D}) = \mathbf{0} \quad (j \neq k),$$

$$\sum_j \ell_j(\mathbf{D}) = \mathbf{I}, \quad \mathbf{D} = \sum_j \lambda_j \ell_j(\mathbf{D}).$$

Conjugation by $\mathbf{S}$ gives the four matrix identities.

For $\mathbf{x} \in \mathcal{E}_{\lambda_k}(\mathbf{A})$,

$$\mathbf{P}_j \mathbf{x} = \ell_j(\lambda_k) \mathbf{x} = \delta_{jk} \mathbf{x}.$$

Hence $\mathcal{E}_{\lambda_j}(\mathbf{A}) \subseteq \text{im } \mathbf{P}_j$. The diagonal representation also gives

$$(\mathbf{A} - \lambda_j \mathbf{I}) \mathbf{P}_j = \mathbf{0},$$

so $\text{im } \mathbf{P}_j \subseteq \mathcal{E}_{\lambda_j}(\mathbf{A})$. 

Exercise 7.8. Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad F_0 = 0, \quad F_1 = 1, \quad F_{k+1} = F_k + F_{k-1} \quad (k \geq 1).$$

Prove, for $k \geq 1$,

$$\mathbf{A}^k = F_k \mathbf{A} + F_{k-1} \mathbf{I}.$$

Deduce $\mathbf{A}^{-1} = \mathbf{A} - \mathbf{I}$.

Origin: Adapted from H Ex. 21.4; S Ex. 3.22

Solution.

$$\chi_{\mathbf{A}}(t) = t^2 - t - 1, \quad \mathbf{A}^2 = \mathbf{A} + \mathbf{I}.$$

The formula holds for $k = 1$. If it holds for k , then

$$\begin{aligned} \mathbf{A}^{k+1} &= F_k \mathbf{A}^2 + F_{k-1} \mathbf{A} \\ &= F_k (\mathbf{A} + \mathbf{I}) + F_{k-1} \mathbf{A} \\ &= F_{k+1} \mathbf{A} + F_k \mathbf{I}. \end{aligned}$$

Finally,

$$\mathbf{A}(\mathbf{A} - \mathbf{I}) = \mathbf{A}^2 - \mathbf{A} = \mathbf{I}.$$



Exercise 7.9. Let $\mathbf{A} \in \mathbb{C}^{n \times n}$ be diagonalisable. Prove that every entry of $\mathbf{A}^k$ tends to 0 as $k \rightarrow \infty$ if and only if every eigenvalue λ of $\mathbf{A}$ satisfies $|\lambda| < 1$.

Origin: Adapted from H Ex. 21.19

Solution. Write

$$\mathbf{A} = \mathbf{S} \operatorname{diag}(\lambda_1, \dots, \lambda_n) \mathbf{S}^{-1}.$$

If $|\lambda_\ell| < 1$ for every ℓ , then

$$(\mathbf{A}^k)_{ij} = \sum_{\ell=1}^n s_{i\ell} \lambda_\ell^k (\mathbf{S}^{-1})_{\ell j} \longrightarrow 0.$$

Conversely, suppose every entry of $\mathbf{A}^k$ tends to 0. For an eigenpair $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$,

$$\mathbf{A}^k \mathbf{x} = \lambda^k \mathbf{x}.$$

Choose i with $x_i \neq 0$. Then

$$\lambda^k x_i = \sum_{j=1}^n (\mathbf{A}^k)_{ij} x_j \longrightarrow 0,$$

so $\lambda^k \rightarrow 0$, which is equivalent to $|\lambda| < 1$. 

Exercise 7.10. Let $\mathbf{A}, \mathbf{B} \in \mathbb{C}^{n \times n}$ satisfy $\mathbf{AB} = \mathbf{BA}$. Suppose $\mathbf{A}$ has n distinct eigenvalues. Prove that there is one nonsingular $\mathbf{S}$ for which both

$$\mathbf{S}^{-1} \mathbf{A} \mathbf{S} \quad \text{and} \quad \mathbf{S}^{-1} \mathbf{B} \mathbf{S}$$

are diagonal.

Origin: Adapted from A&M Ex. 7.39; H Ex. 21.29

Solution. Let $\mathbf{A} \mathbf{x}_j = \lambda_j \mathbf{x}_j$. Commutation gives

$$\mathbf{A}(\mathbf{B} \mathbf{x}_j) = \mathbf{B} \mathbf{A} \mathbf{x}_j = \lambda_j \mathbf{B} \mathbf{x}_j,$$

so $\mathbf{B}\mathbf{x}_j \in \mathcal{E}_{\lambda_j}(\mathbf{A})$. The n roots are distinct, so

$$\dim \mathcal{E}_{\lambda_j}(\mathbf{A}) = g_{\mathbf{A}}(\lambda_j) \leq a_{\mathbf{A}}(\lambda_j) = 1.$$

Since $\mathbf{x}_j \neq \mathbf{0}$ belongs to this eigenspace,

$$\dim \mathcal{E}_{\lambda_j}(\mathbf{A}) = 1, \quad \mathbf{B}\mathbf{x}_j = \mu_j \mathbf{x}_j$$

for some $\mu_j \in \mathbb{C}$, including $\mu_j = 0$. The distinct-eigenspaces theorem makes

$$\mathbf{S} = (\mathbf{x}_1, \dots, \mathbf{x}_n)$$

nonsingular, and

$$\mathbf{S}^{-1}\mathbf{A}\mathbf{S} = \text{diag}(\lambda_1, \dots, \lambda_n), \quad \mathbf{S}^{-1}\mathbf{B}\mathbf{S} = \text{diag}(\mu_1, \dots, \mu_n).$$



Exercise 7.11. Let $\mathbf{A} \in \mathbb{C}^{m \times m}$ and $\mathbf{B} \in \mathbb{C}^{n \times n}$ satisfy

$$\sigma_{\mathbb{C}}(\mathbf{A}) \cap \sigma_{\mathbb{C}}(\mathbf{B}) = \emptyset.$$

Prove that, for every $\mathbf{C} \in \mathbb{C}^{m \times n}$, the equation

$$\mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B} = \mathbf{C}$$

has exactly one solution $\mathbf{X} \in \mathbb{C}^{m \times n}$.

Origin: S Ex. 3.23

Solution. Define

$$\mathcal{L} : \mathbb{C}^{m \times n} \longrightarrow \mathbb{C}^{m \times n}, \quad \mathcal{L}(\mathbf{X}) = \mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B}.$$

Suppose $\mathcal{L}(\mathbf{X}) = \mathbf{0}$. Then

$$\mathbf{A}^k \mathbf{X} = \mathbf{X} \mathbf{B}^k, \quad k \geq 0,$$

by induction. With $p = \chi_{\mathbf{A}}$,

$$p(\mathbf{A})\mathbf{X} = \mathbf{X}p(\mathbf{B}).$$

Cayley–Hamilton gives $p(\mathbf{A}) = \mathbf{O}$. Polynomial spectral mapping gives

$$\sigma_{\mathbb{C}}(p(\mathbf{B})) = \{p(\mu) : \mu \in \sigma_{\mathbb{C}}(\mathbf{B})\}.$$

Every $p(\mu) \neq 0$ because the spectra are disjoint, so $p(\mathbf{B})$ is nonsingular. Hence

$$\mathbf{O} = \mathbf{X}p(\mathbf{B}) \implies \mathbf{X} = \mathbf{O}.$$

Thus $\mathcal{L}$ is injective. Its domain and codomain both have dimension mn , so rank–nullity makes it surjective. Existence and uniqueness follow. ■

Exercise 7.12. Let

$$\mathbf{C} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -1 & 2 \end{pmatrix}, \quad y_k = \mathbf{u}^* \mathbf{C}^k \mathbf{v}$$

for fixed $\mathbf{u}, \mathbf{v} \in \mathbb{C}^3$. Prove

$$y_{k+3} = 2y_{k+2} - y_{k+1} + 4y_k, \quad k \geq 0.$$

Origin: Adapted from A&M Ex. 7.38

Solution.

$$\begin{aligned} \chi_{\mathbf{C}}(t) &= \det \begin{pmatrix} t & -1 & 0 \\ 0 & t & -1 \\ -4 & 1 & t-2 \end{pmatrix} \\ &= t\{t(t-2) + 1\} - 4 = t^3 - 2t^2 + t - 4. \end{aligned}$$

Cayley–Hamilton and multiplication by $\mathbf{C}^k$ give

$$\mathbf{C}^{k+3} - 2\mathbf{C}^{k+2} + \mathbf{C}^{k+1} - 4\mathbf{C}^k = \mathbf{0}.$$

Premultiplication by $\mathbf{u}^*$ and postmultiplication by $\mathbf{v}$ give

$$y_{k+3} - 2y_{k+2} + y_{k+1} - 4y_k = 0.$$



Exercise 7.13. Let $a_0, a_1, a_2, a_3 \in \mathbb{C}$ and

$$\mathbf{P} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad p(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3, \quad \mathbf{C} = p(\mathbf{P}).$$

For $\omega \in \{1, i, -1, -i\}$, set

$$\mathbf{f}_\omega = (1 \ \omega^{-1} \ \omega^{-2} \ \omega^{-3})'.$$

(a) Prove that the four $\mathbf{f}_\omega$ form a common eigenbasis for every matrix $p(\mathbf{P})$ and that

$$\mathbf{C}\mathbf{f}_\omega = p(\omega)\mathbf{f}_\omega.$$

(b) Prove

$$\det \mathbf{C} = p(1)p(i)p(-1)p(-i).$$

(c) If the four factors are nonzero, construct a polynomial q of degree at most 3 such that $\mathbf{C}^{-1} = q(\mathbf{P})$.

Solution. Since $\omega^4 = 1$,

$$\mathbf{P}\mathbf{f}_\omega = \begin{pmatrix} \omega^{-3} \\ 1 \\ \omega^{-1} \\ \omega^{-2} \end{pmatrix} = \omega \mathbf{f}_\omega.$$

The four eigenvalues of $\mathbf{P}$ are distinct, so the four eigenvectors are linearly independent. Polynomial action gives

$$p(\mathbf{P})\mathbf{f}_\omega = p(\omega)\mathbf{f}_\omega.$$

Thus $\mathbf{C}$ is similar to the diagonal matrix with diagonal $p(1), p(i), p(-1), p(-i)$, proving the determinant formula.

If every $p(\omega) \neq 0$, define

$$q(t) = \sum_{\omega \in \{1, i, -1, -i\}} \frac{1}{p(\omega)} \prod_{\substack{\zeta \in \{1, i, -1, -i\} \\ \zeta \neq \omega}} \frac{t - \zeta}{\omega - \zeta}.$$

Then $\deg q \leq 3$ and $q(\omega)p(\omega) = 1$. Hence

$$q(\mathbf{P})\mathbf{C}\mathbf{f}_\omega = \mathbf{f}_\omega$$

on a basis, so $q(\mathbf{P})\mathbf{C} = \mathbf{I}$ and $q(\mathbf{P}) = \mathbf{C}^{-1}$. ■

Exercise 7.14. Let $\mathbf{T}$ be upper triangular with diagonal $\lambda_1, \dots, \lambda_n$, and set

$$\mathbf{B}_j = \mathbf{T} - \lambda_j \mathbf{I}, \quad \mathbf{P}_k = \mathbf{B}_1 \cdots \mathbf{B}_k.$$

Prove by induction that the leading $k \times k$ block of $\mathbf{P}_k$ is zero. Deduce

$$\prod_{j=1}^n (\mathbf{T} - \lambda_j \mathbf{I}) = \mathbf{O}.$$

Use Schur triangularisation to obtain an alternative proof of Cayley–Hamilton for every complex square matrix.

Origin: A&M Exs. 7.56 and 7.69

Solution. For $k = 1$, $(\mathbf{P}_1)_{11} = (\mathbf{B}_1)_{11} = 0$. Suppose the leading $(k-1) \times (k-1)$ block of $\mathbf{P}_{k-1}$ is zero. Since all $\mathbf{B}_j$ are polynomials in $\mathbf{T}$, they commute, and

$$\mathbf{P}_k = \mathbf{P}_{k-1}\mathbf{B}_k.$$

For $i, j \leq k-1$, upper-triangular multiplication uses only indices $i \leq \ell \leq j$, so

$$(\mathbf{P}_k)_{ij} = \sum_{\ell=i}^j (\mathbf{P}_{k-1})_{i\ell} (\mathbf{B}_k)_{\ell j} = 0.$$

For $i < k$,

$$(\mathbf{P}_k)_{ik} = \sum_{\ell=i}^{k-1} (\mathbf{P}_{k-1})_{i\ell} (\mathbf{B}_k)_{\ell k} + (\mathbf{P}_{k-1})_{ik} (\mathbf{B}_k)_{kk} = 0,$$

because the first terms lie in the zero leading block and $(\mathbf{B}_k)_{kk} = 0$. Also $(\mathbf{P}_k)_{ki} = 0$ for $i < k$, and triangular multiplication gives

$$(\mathbf{P}_k)_{kk} = \prod_{r=1}^k (\lambda_k - \lambda_r) = 0.$$

The induction is complete. At $k = n$,

$$\mathbf{P}_n = \prod_{j=1}^n (\mathbf{T} - \lambda_j \mathbf{I}) = \mathbf{O}.$$

For $\mathbf{A} = \mathbf{Q}\mathbf{T}\mathbf{Q}^*$ from Schur's theorem,

$$\chi_{\mathbf{A}}(t) = \chi_{\mathbf{T}}(t) = \prod_{j=1}^n (t - \lambda_j).$$

Matrix-polynomial algebra gives

$$\chi_{\mathbf{A}}(\mathbf{A}) = \mathbf{Q}\chi_{\mathbf{T}}(\mathbf{T})\mathbf{Q}^* = \mathbf{O}.$$

